STATS 506 Homework4

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This is the git link : https://github.com/wei-min-lin/Homework1/tree/main/Homework2

```
# If needed, run once in the R Console:
# install.packages(c("dplyr", "ggplot2", "knitr", "scales"))
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(knitr)

dir.create("data", showWarnings = FALSE)
urls <- c(
  "nzge.rds" = "https://phuongov.github.io/stats506-f26/data_f26/nzge.rds",
  "atp_matches_2019.csv" = "https://raw.githubusercontent.com/azhou5849/tennis-atp-sackmann-data/master/at
  "us-states.csv" = "https://raw.githubusercontent.com/nytimes/covid-19-data/refs/heads/master/rolling-ave
)
for (filename in names(urls)) {
  destination <- file.path("data", filename)
  if (!file.exists(destination)) {
    download.file(urls[[filename]], destination, mode = "wb", quiet = TRUE)
  }
}

# Data loading
nzge <- readRDS("data/nzge.rds")
tennis <- read.csv("data/atp_matches_2019.csv")
covid <- read.csv("data/us-states.csv") |>
  mutate(date = as.Date(date)) |>
  filter(state %in% state.name) |>
  arrange(state, date)

theme_set(theme_minimal(base_size = 12))
theme_update(panel.grid.minor = element_blank(),
             plot.title = element_text(face = "bold"),
             legend.position = "bottom")
```

Problem 1(a)

```
votes_by_year <- nzge |>
  group_by(election_year, voting_type) |>
  summarise(votes = sum(votes, na.rm = TRUE), .groups = "drop") |>
  arrange(desc(votes))
kable(votes_by_year)
```

election_year	voting_type	votes
2014	Party	2416479
2014	Candidate	2375493
2008	Party	2356536
2008	Candidate	2325598
2005	Party	2286190
2005	Candidate	2260670
2011	Party	2257336
2011	Candidate	2225766
2002	Party	2040248
2002	Candidate	2022115

Problem 1(b)

Percentages use all recorded Candidate votes as the denominator. Informal votes are shown explicitly; they are not a political party.

```
# Informal votes are shown explicitly and are not interpreted as a political party.
candidate_2014 <- nzge |>
  filter(election_year == 2014, voting_type == "Candidate") |>
  group_by(party) |>
  summarise(votes = sum(votes, na.rm = TRUE), .groups = "drop") |>
  mutate(percent = 100 * votes / sum(votes)) |>
  arrange(desc(percent))
kable(candidate_2014, digits = 2)
```

party	votes	percent
National Party	1081787	45.54
Labour Party	801287	33.73
Green Party	165718	6.98
Conservative Party	81075	3.41
New Zealand First Party	73384	3.09
Maori Party	42108	1.77
MANA Movement	32333	1.36
Informal Candidate Votes	27886	1.17
ACT New Zealand	27778	1.17
United Future	14722	0.62
Aotearoa Legalise Cannabis Party	4936	0.21
Internet Party	4848	0.20
Democrats for Social Credit	4647	0.20
Ban1080	4448	0.19
Independent	3864	0.16
NZ Independent Coalition	1929	0.08
Focus New Zealand	1797	0.08
Money Free Party	391	0.02
Communist League	135	0.01
Climate Party	116	0.00

party	votes	percent
Human Rights Party	76	0.00
The Expats	70	0.00
Alliance	59	0.00
NZ Economic Euthenics Party	51	0.00
Patriotic Revolutionary Front	48	0.00

Problem 1(c)

The winner is the party with the greatest national vote total for each vote type, not necessarily the most seats.

```
election_winners <- nzge |>
  filter(!grepl("^Informal", party)) |>
  group_by(election_year, voting_type, party) |>
  summarise(votes = sum(votes, na.rm = TRUE), .groups = "drop") |>
  group_by(election_year, voting_type) |>
  slice_max(votes, n = 1, with_ties = TRUE) |>
  ungroup() |>
  group_by(election_year) |>
  summarise(
    Candidate = paste(party[voting_type == "Candidate"], collapse = "; "),
    Party = paste(party[voting_type == "Party"], collapse = "; "),
    .groups = "drop"
  ) |>
  arrange(election_year)
kable(election_winners)
```

election_year	Candidate	Party
2002	Labour Party	Labour Party
2005	National Party	Labour Party
2008	National Party	National Party
2011	National Party	National Party
2014	National Party	National Party

Problem 2(a)

A tournament is defined by a distinct `tourney_id`, including Davis Cup ties.

```
tournament_count <- tennis |>
  summarise(tournaments = n_distinct(tourney_id))
kable(tournament_count)
```

tournaments
128

```
cat(sprintf("The dataset contains %d distinct tournament IDs, including Davis Cup ties.\n",
            tournament_count$tournaments))
```

```
The dataset contains 128 distinct tournament IDs, including Davis Cup ties.
```

Problem 2(b)

Individual singles titles are identified by final-round wins; Davis Cup team ties are excluded.

```r
titles <- tennis |>
  filter(round == "F", tourney_level != "D") |>
  distinct(tourney_id, winner_id, winner_name) |>
  count(winner_id, winner_name, name = "titles", sort = TRUE)
multiple_titles <- titles |> filter(titles > 1)
top_champions <- titles |> slice_max(titles, n = 1, with_ties = TRUE)
kable(multiple_titles)
```

winner_id	winner_name	titles
104925	Novak Djokovic	5
106233	Dominic Thiem	5
103819	Roger Federer	4
104745	Rafael Nadal	4
106421	Daniil Medvedev	4
126774	Stefanos Tsitsipas	3
200282	Alex De Minaur	3
104542	Jo-Wilfried Tsonga	2
105332	Benoit Paire	2
106401	Nick Kyrgios	2
106426	Cristian Garin	2
126610	Matteo Berrettini	2

```r
kable(top_champions)
```

winner_id	winner_name	titles
104925	Novak Djokovic	5
106233	Dominic Thiem	5

```r
cat(sprintf("%d players won more than one singles tournament. %s each won %d titles.\n",
            nrow(multiple_titles), paste(top_champions$winner_name, collapse = " and "),
            max(titles$titles)))
```

```
12 players won more than one singles tournament. Novak Djokovic and Dominic Thiem each won 5 titles.
```

Problem 2(c)

This is a descriptive comparison of winner and loser ace counts within the same match. No hypothesis test is used.

```r
ace_pairs <- tennis |>
  filter(!is.na(w_ace), !is.na(l_ace)) |>
  mutate(difference = w_ace - l_ace)
ace_summary <- ace_pairs |>
  summarise(
    complete_matches = n(),
    winner_mean = mean(w_ace), loser_mean = mean(l_ace),
    mean_difference = mean(difference), median_difference = median(difference),
    winner_more_percent = 100 * mean(difference > 0),
    equal_percent = 100 * mean(difference == 0),
    loser_more_percent = 100 * mean(difference < 0)
  )
kable(ace_summary, digits = 2)
```

complete_matches	winner_mean	loser_mean	mean_difference	median_difference	winner_more_percent	equal_percent	loser_more_percent
2694	7.5	5.79	1.7	1	57.28	8.35	34.37

```
cat(sprintf(paste0("In %d complete pairs, winners averaged %.2f aces and losers %.2f; ",
                   "the mean paired difference was %.2f. Winners had more aces in %.1f%% ",
                   "of matches. This is descriptive evidence of an association, not ",
                   "a causal conclusion or a formal significance test. %d matches ",
                   "with missing ace statistics were excluded.\n"),
            nrow(ace_pairs), ace_summary$winner_mean, ace_summary$loser_mean,
            ace_summary$mean_difference, ace_summary$winner_more_percent,
            nrow(tennis) - nrow(ace_pairs)))
```

```
In 2694 complete pairs, winners averaged 7.50 aces and losers 5.79; the mean paired difference was 1.70. W
```

Problem 2(d)

```
player_records <- bind_rows(
  tennis |> transmute(player_id = winner_id, player = winner_name, win = 1L),
  tennis |> transmute(player_id = loser_id, player = loser_name, win = 0L)
) |>
  group_by(player_id) |>
  summarise(player = first(player), matches = n(), wins = sum(win),
            .groups = "drop") |>
  filter(matches >= 5) |>
  mutate(win_rate = wins / matches)
best_players <- player_records |> slice_max(win_rate, n = 1, with_ties = TRUE)
kable(best_players |> mutate(win_percent = 100 * win_rate) |> select(-win_rate),
      digits = 2)
```

player_id	player	matches	wins	win_percent
104745	Rafael Nadal	69	60	86.96

```
cat(sprintf("Among players with at least five matches, %s had the highest win rate: %.2f%%.\n",
            paste(best_players$player, collapse = "; "),
            100 * max(best_players$win_rate)))
```

```
Among players with at least five matches, Rafael Nadal had the highest win rate: 86.96%.
```

Problem 3(a)

Only the 50 states are included. Major and minor peaks are operational definitions, not official epidemiological classifications.

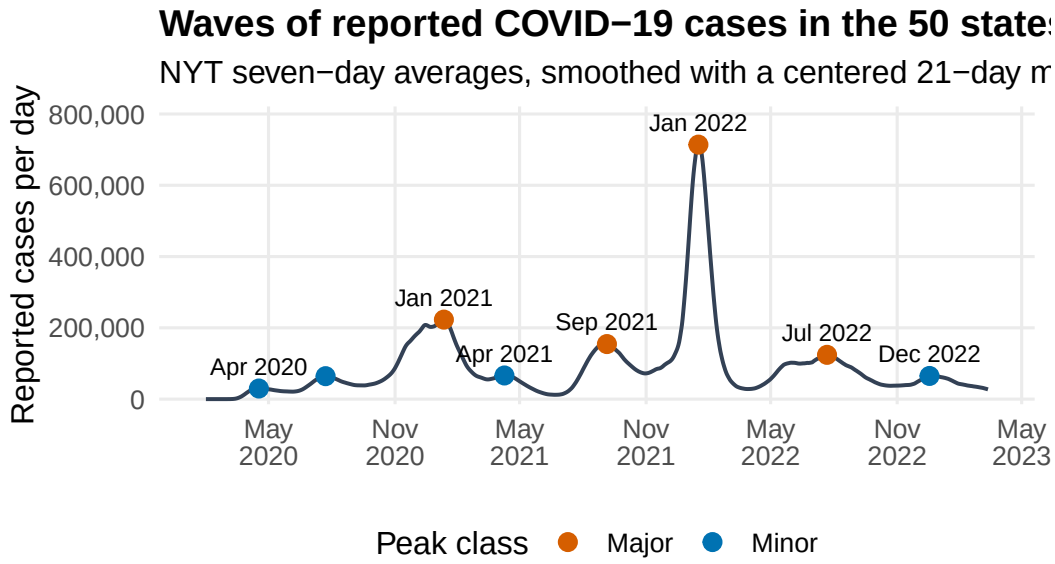
```
#' Identify separated local peaks in a smoothed daily series.
#' @param y Numeric daily values, including possible NA edge values.
#' @param window Days examined on either side of each candidate peak.
#' @param minimum Minimum peak height.
#' @param separation Minimum distance in days between accepted peaks.
#' @return Integer indices of accepted peaks in chronological order.
find_peaks <- function(y, window = 30L, minimum = 20000, separation = 60L) {
  candidates <- integer()
  for (i in seq.int(window + 1L, length(y) - window)) {
    neighborhood <- y[(i - window):(i + window)]
    if (!anyNA(neighborhood) && y[i] >= minimum &&
```

```
        y[i] == max(neighborhood)) candidates <- c(candidates, i)
  }
  accepted <- integer()
  for (i in candidates[order(y[candidates], decreasing = TRUE)]) {
    if (all(abs(i - accepted) >= separation)) accepted <- c(accepted, i)
  }
  sort(accepted)
}

national <- covid |>
  group_by(date) |>
  summarise(cases_avg = sum(cases_avg, na.rm = TRUE), .groups = "drop") |>
  arrange(date) |>
  mutate(smooth = as.numeric(stats::filter(cases_avg, rep(1 / 21, 21), sides = 2)))
peaks <- national |>
  slice(find_peaks(smooth)) |>
  mutate(type = if_else(smooth >= 100000, "Major", "Minor"))
kable(peaks |> select(date, smooth, type), digits = 0)
```

date	smooth	type
2020-04-17	29736	Minor
2020-07-23	64412	Minor
2021-01-11	223061	Major
2021-04-09	66329	Minor
2021-09-05	154471	Major
2022-01-16	714052	Major
2022-07-22	124259	Major
2022-12-18	65015	Minor

```
p3a <- ggplot(national, aes(date, smooth)) +
  geom_line(color = "#334155", linewidth = 0.7, na.rm = TRUE) +
  geom_point(data = peaks, aes(color = type), size = 2.8) +
  geom_text(data = peaks, aes(label = format(date, "%b %Y")),
            vjust = -0.8, size = 3, check_overlap = TRUE) +
  scale_color_manual(values = c(Major = "#D55E00", Minor = "#0072B2")) +
  scale_y_continuous(labels = scales::label_comma(),
                     expand = expansion(mult = c(0.02, 0.15))) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
  labs(title = "Waves of reported COVID-19 cases in the 50 states",
       subtitle = "NYT seven-day averages, smoothed with a centered 21-day mean",
       x = NULL, y = "Reported cases per day", color = "Peak class",
       caption = "Source: The New York Times. Major >=100,000; minor 20,000-99,999.\nLocal maxima within +
print(p3a)
```




```
cat(sprintf("Under this operational definition, the plot contains %d major and %d minor peaks. Counts depe
            sum(peaks$type == "Major"), sum(peaks$type == "Minor")))
```

```
Under this operational definition, the plot contains 4 major and 4 minor peaks. Counts depend on the smoot
```

Problem 3(b)

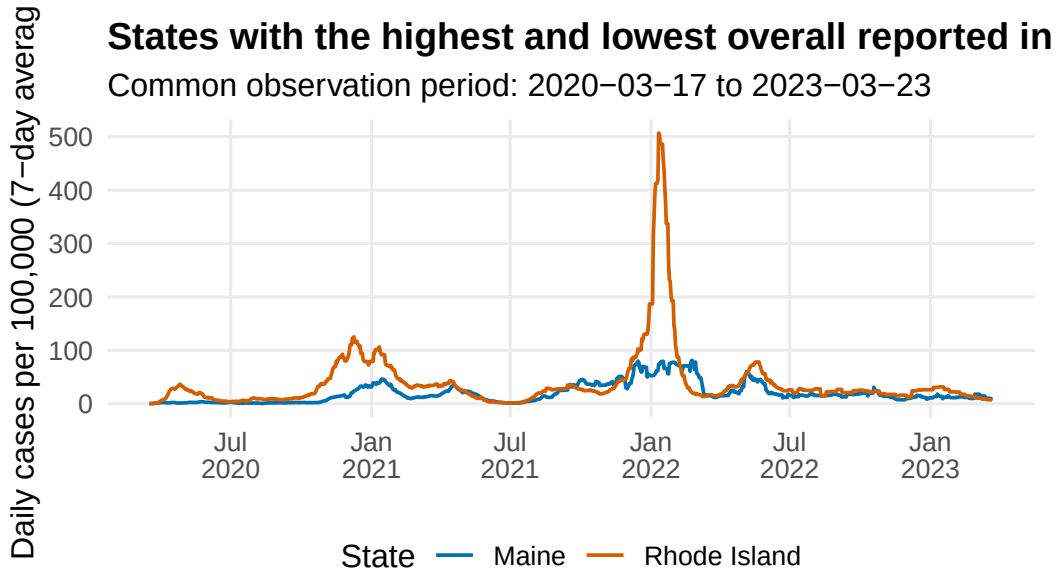
Overall rate means average daily reported cases per 100,000 over the common observation period. Equal observation periods make these rates comparable.

```
coverage <- covid |>
  group_by(state) |>
  summarise(first_day = min(date), last_day = max(date), .groups = "drop")
common_start <- max(coverage$first_day)
common_end <- min(coverage$last_day)
common <- covid |> filter(date >= common_start, date <= common_end)
state_rates <- common |>
  group_by(state) |>
  summarise(overall_rate = mean(cases_avg_per_100k), .groups = "drop")
extreme_states <- bind_rows(
  state_rates |> slice_max(overall_rate, n = 1, with_ties = TRUE) |> mutate(group = "Highest"),
  state_rates |> slice_min(overall_rate, n = 1, with_ties = TRUE) |> mutate(group = "Lowest")
)
kable(extreme_states, digits = 2)
```

state	overall_rate	group
Rhode Island	39.29	Highest
Maine	19.98	Lowest

```
comparison <- common |> semi_join(extreme_states, by = "state")
p3b <- ggplot(comparison, aes(date, cases_avg_per_100k, color = state)) +
  geom_line(linewidth = 0.65) +
  scale_color_manual(values = c("#0072B2", "#D55E00")) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
```

```
  labs(title = "States with the highest and lowest overall reported incidence",
       subtitle = paste("Common observation period:", common_start, "to", common_end),
       x = NULL, y = "Daily cases per 100,000 (7-day average)", color = "State",
       caption = "Source: The New York Times. Overall rate = mean daily 7-day incidence.\nReported cases d
print(p3b)
```



Source: The New York Times. Overall rate = mean daily 7−day incidence.
Reported cases depend on testing and reporting practices.

```
trajectory_summary <- comparison |>
  group_by(state) |>
  summarise(
    peak_rate = max(cases_avg_per_100k),
    peak_date = date[which.max(cases_avg_per_100k)],
    days_above_50 = sum(cases_avg_per_100k >= 50), .groups = "drop"
  )
kable(trajectory_summary, digits = 2)
```

state	peak_rate	peak_date	days_above_50
Maine	81.21	2022-02-23	99
Rhode Island	506.81	2022-01-11	189

```
cat(sprintf("%s had the highest overall rate and %s the lowest.\n",
            paste(extreme_states$state[extreme_states$group == "Highest"], collapse = "; "),
            paste(extreme_states$state[extreme_states$group == "Lowest"], collapse = "; ")))
```

```
Rhode Island had the highest overall rate and Maine the lowest.
```

```
cat(paste(sprintf("%s peaked at %.2f daily cases per 100,000 on %s and spent %d days at or above 50.",
                  trajectory_summary$state, trajectory_summary$peak_rate,
                  trajectory_summary$peak_date, trajectory_summary$days_above_50),
          collapse = " "),
    "The higher-rate state had a much taller peak and a longer period of high incidence; the peaks also oc
```

```
Maine peaked at 81.21 daily cases per 100,000 on 2022-02-23 and spent 99 days at or above 50. Rhode Island
```

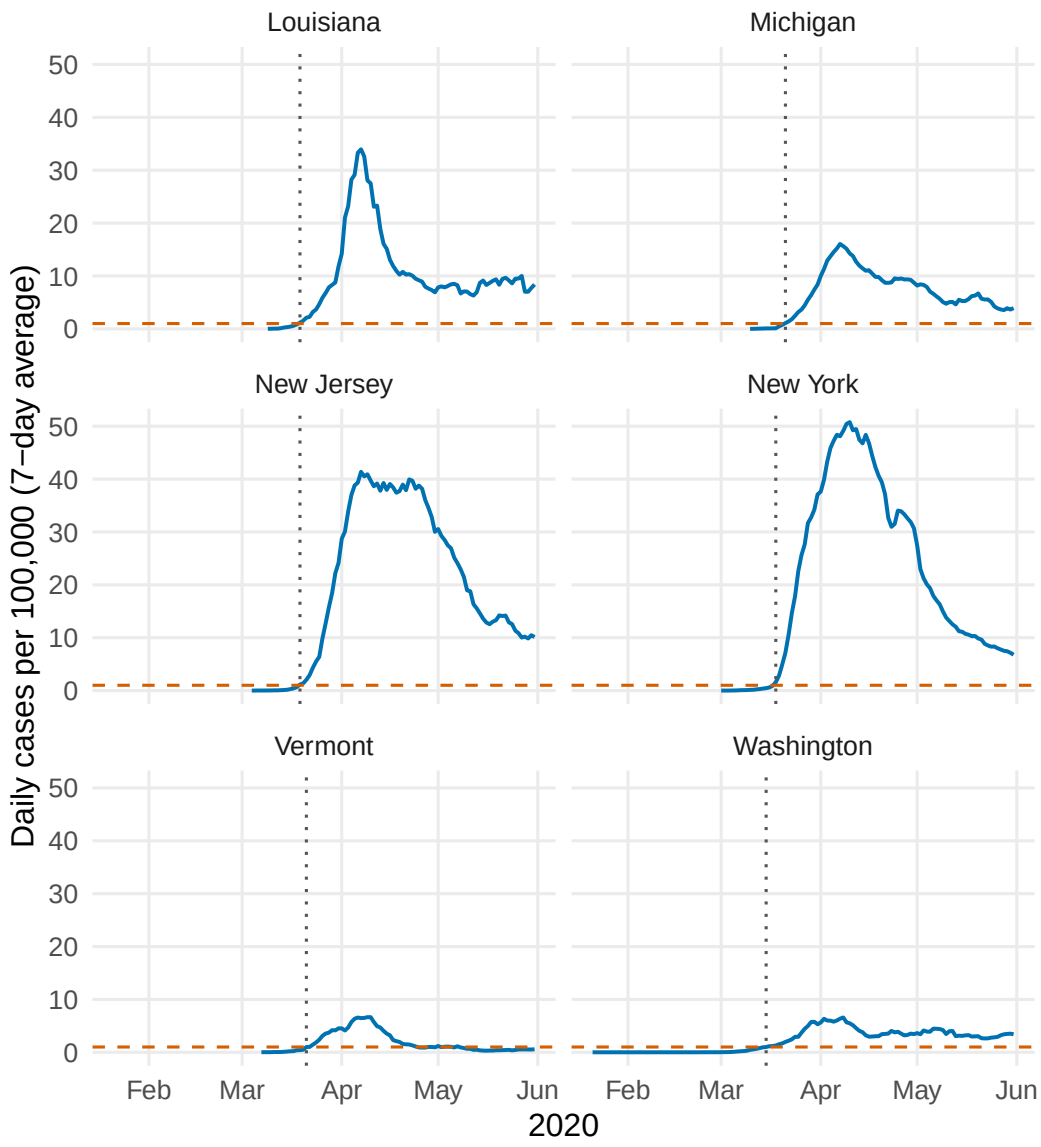
Problem 3(c)

Substantial incidence means at least one daily reported case per 100,000 (7-day average) for seven consecutive days. Ties at fifth place are retained.

```
early <- covid |>
  filter(date <= as.Date("2020-05-31")) |>
  group_by(state) |>
  arrange(date, .by_group = TRUE) |>
  mutate(sustained = as.numeric(stats::filter(
    as.integer(cases_avg_per_100k >= 1), rep(1, 7), sides = 1
  )) == 7) |>
  ungroup()
onsets <- early |>
  filter(sustained) |>
  group_by(state) |>
  summarise(onset = min(date) - 6, .groups = "drop") |>
  arrange(onset, state)
first_states <- onsets |> slice_min(onset, n = 5, with_ties = TRUE)
kable(first_states)
```

state	onset
Washington	2020-03-15
New York	2020-03-18
Louisiana	2020-03-19
New Jersey	2020-03-19
Michigan	2020-03-21
Vermont	2020-03-21

```
p3c <- early |>
  semi_join(first_states, by = "state") |>
  ggplot(aes(date, cases_avg_per_100k)) +
  geom_line(color = "#0072B2", linewidth = 0.7) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "#D55E00") +
  geom_vline(data = first_states, aes(xintercept = onset),
             linetype = "dotted", color = "#555555") +
  facet_wrap(~state, ncol = 2) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Early sustained COVID-19 outbreaks",
       subtitle = "First five states; any ties at fifth place are retained",
       x = "2020", y = "Daily cases per 100,000 (7-day average)",
       caption = "Source: The New York Times. Dashed line: 1 case per 100,000.\nDotted lines: start of the
print(p3c)
```




```
cat(sprintf("Using the stated threshold, the earliest states were %s. These are dates of sustained reporte
            paste(first_states$state, collapse = ", ")))
```

```
Using the stated threshold, the earliest states were Washington, New York, Louisiana, New Jersey, Michigan
```